

Customer Segmentation using K-Means Clustering

1. Introduction

Customer segmentation is an important technique used in data analytics to divide customers into different groups based on their behavior. Businesses use segmentation to better understand customer needs and improve marketing strategies.

In this project, clustering techniques were applied to a customer dataset to identify different groups of customers based on their purchasing behavior and financial activity.

2. Dataset Description

The dataset used for this analysis contains information about **8950 credit card customers** with **18 variables** describing their financial behavior.

Some important variables include:

- **BALANCE** – Amount of balance in the account
- **PURCHASES** – Total purchase amount
- **ONEOFF_PURCHASES** – Purchases made in a single transaction
- **INSTALLMENTS_PURCHASES** – Purchases made through installments
- **CASH_ADVANCE** – Cash withdrawn using credit card
- **CREDIT_LIMIT** – Credit limit assigned to the customer
- **PAYMENTS** – Total payments made by the customer
- **TENURE** – Duration of the credit card usage

These variables help understand the spending habits and financial behavior of customers.

3. Data Preprocessing

Before applying clustering, several preprocessing steps were performed:

1. **Handling missing values** in the dataset.
2. **Removing the customer ID column** since it does not contribute to clustering.
3. **Feature scaling (normalization)** was applied to ensure that variables with larger values do not dominate the clustering process.
4. The dataset was prepared for clustering by converting all variables into numerical format suitable for machine learning algorithms.

4. Clustering Method

The **K-Means clustering algorithm** was used to group customers into different segments.

K-Means works by:

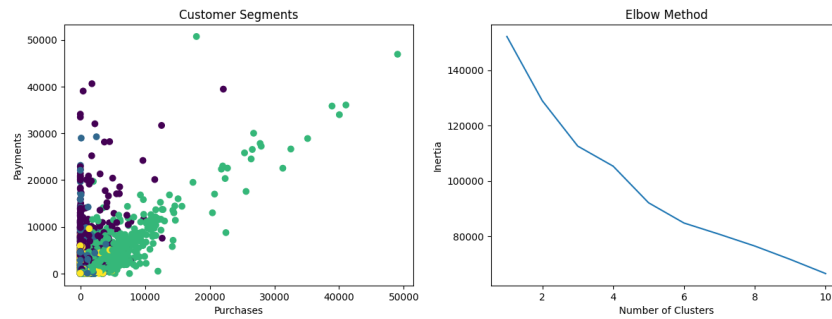
1. Selecting a number of clusters (K).
2. Assigning each customer to the nearest cluster center.
3. Updating cluster centers iteratively until the clusters stabilize.

The optimal number of clusters was selected based on clustering analysis techniques such as the **Elbow Method**.

5. Results and Visualization

After applying the K-Means algorithm, customers were grouped into different clusters based on their financial and purchasing behavior.

The visualization of clusters helped identify patterns among customers. Each cluster represents a group of customers with similar characteristics.



Examples of customer segments may include:

Cluster 1 – High Spenders

- High purchase frequency
- Large credit limits
- Higher payment amounts

Cluster 2 – Moderate Users

- Moderate purchases
- Average credit limits
- Regular payments

Cluster 3 – Low Activity Customers

- Low purchases
- Low balance
- Limited credit card usage

Visualizations such as scatter plots and cluster graphs were used to clearly show the separation between customer segments.

6. Conclusion

Customer segmentation using K-Means clustering helped identify different groups of customers based on their purchasing behavior. The analysis revealed distinct customer segments with varying financial patterns.

These insights can help businesses design better marketing strategies, offer personalized services, and improve customer relationship management.

Clustering techniques like K-Means are powerful tools in data analytics for discovering hidden patterns in large datasets.